

CHE 313 Transport Phenomena III
Homework #5 (Due:11/13/25, 1:00 pm)
TA office hour (T:10am-12pm; TR:10am-11am): Aelx Acquah (aacqua2@uic.edu)

1. Textbook (Wankat, P. C. 5th edition), 9.D1
2. Textbook (Wankat, P. C. 5th edition), 9.D3
3. Textbook (Wankat, P. C. 5th edition), 9.D6
4. Textbook (Wankat, P. C. 5th edition), 9.D11
5. Textbook (Wankat, P. C. 5th edition), 9.D16

W_{feed} distillate D_{total} and x_{D,avg}. VLE data are in Table 2-8. p = 1.0 atm.

a. Use Simpson's rule.

b. Use the Gaussian quadrature formula.

2

$$\frac{x_F + x_f}{2}$$

$$W_F = F e^{-\text{area}}$$

$$D_{\text{tot}} = F - W_F$$

$$x_{D,\text{avg}} = \frac{F x_F - W_F x_F}{D_{\text{tot}}}$$

from notes

X _D	y	$\frac{1}{y-x}$
0.7	0.87	5.88
0.4	0.724	3.04
0.1	0.418	3.14

$$F = 0.6 \text{ kmol}$$

$$x_F = 0.7$$

$$x_f = 0.10$$

Simpson's rule

$$\frac{0.7-0.1}{6} [3.14 + 4(3.04) + 5.88] = 0.072 \text{ kmol}$$

$$W_F = F e^{-0.072} = 0.072 \text{ kmol}$$

$$D_{\text{tot}} = 0.6 - 0.072 = 0.528$$

$$= 0.528 \text{ kmol}$$

b)

$$h = \frac{(x_F - x_f)}{2} = 0.3$$

$$\text{Area} = \frac{h}{9} \left[5 \cdot (0.4) - h\sqrt{0.6} + 8(0.4) + 5(0.4 + h\sqrt{0.6}) \right]$$

$$5 \cdot 0.1676 + 8(0.4) + 5 \cdot 0.6324$$

$$\text{Area} = 2.064$$

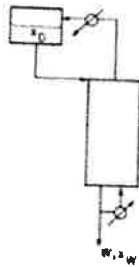
$$W_F = 0.6 e^{-2.064}$$

$$= 0.076 \text{ kmol}$$

$$D_{\text{tot}} = 0.6 - 0.076 = 0.524 \text{ kmol}$$

$$x_{D,\text{avg}} = \frac{(0.6)(0.7) - (0.076)(0.1)}{0.524} = 0.79$$

10. A distillation system with a still pot plus a column with one equilibrium stage is used to batch distill 1.0 kmol of a 57.0 mol% methanol and 43.0 mol% water feed. A total condenser is used. The final bottoms concentration is 15.0 mol% methanol. Pressure is 101.3 kPa. Reflux is a saturated liquid, and $L/D = 1.85$. Find W_{ind} , D_{tot} , and $X_{D, \text{avg}}$. VLE data are in Table 2-8.



$$X_F = 0.57$$

$$X_F = 1.0 \text{ kmol}$$

$$X_B = 0.15$$

$$\frac{L}{D} = 1.85$$

$$\frac{L}{D} = \frac{1.85}{11.85}$$

$$y = 0.65x + 0.35x_D$$

x_0	x_w	$x_0 - x_w$	$\frac{1}{x_0 - x_w}$
0.95	0.47	0.48	2.08
0.90	0.38	0.52	1.92
x_{ind} 0.85	0.37	0.63	2.099
x_f 0.80	0.27	0.53	1.905
0.75	0.13	0.62	1.61
0.70	0.09	0.60	1.66
0.65	0.07	0.57	1.75

$$x_{\text{ind}} = \frac{x_f + x_D}{2} = 0.81$$

Simpson Rule

$$A = \frac{0.57 - 0.15}{6} [1.905 + 4(2.08) + 2.93] = 0.924$$

$$W_F = (110) e^{-0.926} = 39.6 \text{ kmol}$$

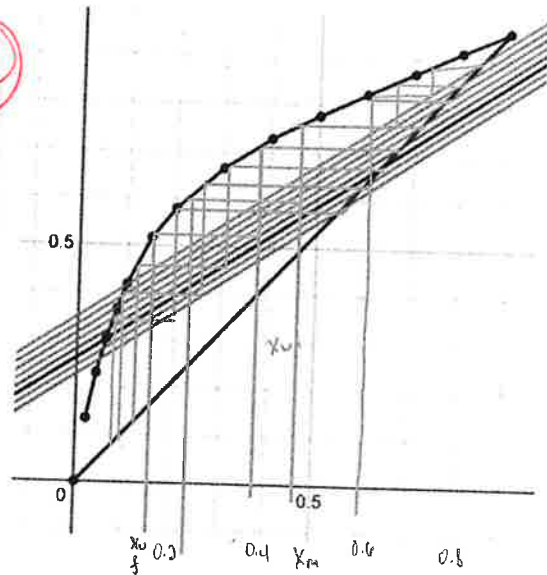
$$F = W_F + D_{\text{tot}}$$

$$D_{\text{tot}} = F - W_F$$

$$D_{\text{tot}} = 60.4 \text{ kmol}$$

$$X_{D, \text{avg}} = \frac{F X_F - W_F \cdot X_F}{D_{\text{tot}}}$$

$$= \frac{110(0.57) - (39.6)(0.15)}{60.4} = 0.845$$



x	y
0.06	0.301
0.08	0.365
0.10	0.418
0.15	0.517
0.20	0.579
0.30	0.665
0.40	0.729
0.50	0.779
0.60	0.825
0.70	0.870
0.80	0.915
0.90	0.958
1.0	1.0

$$y = 0.65x + (0.35 \cdot 0.90)$$

$$y = 0.65x + (0.35 \cdot 0.85)$$

$$y = 0.65x + (0.35 \cdot 0.80)$$

$$y = 0.65x + (0.35 \cdot 0.75)$$

$$y = 0.65x + (0.35 \cdot 0.70)$$

$$y = 0.65x + (0.35 \cdot 0.65)$$

$$y = 0.65x + (0.35 \cdot 0.60)$$

$$y = 0.65x + (0.35 \cdot 0.55)$$

D6. A simple batch still is separating a feed that is 60.0 mol% 1,2-dichloroethane and 40.0 mol% 1,1,2-trichloroethane. Pressure is 1 atm. The relative volatility is constant, $\alpha_{2,1} = 2.4$. The use of Eq. (9-13) is recommended.

- The charge (F) is 1.3 kmol. The final still pot concentration is 30.0 mol% 1,2-dichloroethane. Find the final moles in the still pot and the average mole fraction of distillate product.
- Repeat part a if $F = 3.5$ kmol.
- If $F = 2.0$ kmol and the average distillate concentration is 75.0 mol% 1,2-dichloroethane, find final moles in the still pot and the final mole fraction of 1,2-dichloroethane in the still pot.

Eqn 9-13: $\ln\left(\frac{W_F}{F}\right) = \frac{1}{\alpha-1} \ln\left(\frac{x_f(1-x_F)}{x_F(1-x_f)}\right) + \ln\left(\frac{1-x_F}{1-x_f}\right)$

a) $F = 1.3 \text{ kmol}$ $x_F = 0.6$ $W_f = ?$
 $x_f = 0.30$ $\alpha_{2,1} = 2.4$

$$\ln\left(\frac{W_F}{F}\right) = \frac{1}{2.4-1} \ln\left(\frac{0.3(1-0.6)}{0.6(1-0.3)}\right) + \ln\left(\frac{1-0.6}{1-0.3}\right)$$

$$\ln\left(\frac{W_F}{1.3}\right) = -1.454$$

$$W_F = e^{-1.454}$$

$$W_F = 0.3037 \text{ kmol}$$

$$D_{\text{tot}} = 1.3 - 0.3037 = 0.9963 \text{ kmol}$$

$$F = 3.5 \text{ kmol} \quad x_{D,\text{avg}} = \frac{F \cdot x_F - W_F \cdot x_f}{D_{\text{tot}}} = \frac{(1.3)(0.6) - (0.3037)(0.3)}{0.9963} = 0.6914$$

b) $\ln\left(\frac{W_F}{F}\right) = \ln\left(\frac{W_F}{3.5}\right) = -1.454$ (calculated from a)

$$W_F = 0.817 \text{ kmol}$$

$$D_{\text{tot}} = 3.5 - 0.817 = 2.681 \text{ kmol}$$

$$x_{D,\text{avg}} = \frac{(3.5)(0.6) - (0.817)(0.3)}{2.681} = 0.6916$$

c) $F = 2$

$$x_F = 0.6$$

$$x_{D,\text{avg}} = 0.75$$

$$D_{\text{tot}} x_{D,\text{avg}} = F x_F - W_F x_f$$

$$(F \cdot W_F) x_{D,\text{avg}} = F x_F - W_F x_f$$

$$F x_{D,\text{avg}} - W_F x_{D,\text{avg}} = F x_F - W_F x_f$$

$$\ln\left(\frac{x_{D,\text{avg}} - x_F}{x_{D,\text{avg}} - x_f}\right) = \frac{1}{\alpha-1} \ln\left[\frac{x_f(1-x_F)}{x_F(1-x_f)}\right] + \ln\left(\frac{1-x_F}{1-x_f}\right)$$

$$\ln \left(\frac{0.75 - 0.4}{0.75 - x_f} \right) = \frac{1}{1.4} \ln \left[\frac{0.4 x_f}{0.6(1-x_f)} \right] + \ln \left[\frac{0.4}{1-x_f} \right] \rightarrow e^{\ln \left(\frac{0.15}{0.75-x_f} \right)} = e^{\ln \left[\left(\frac{0.4 x_f}{0.6(1-x_f)} \right)^{1/1.4} \cdot \frac{0.4}{1-x_f} \right]}$$

$$\ln \left(\frac{0.75 - 0.6}{0.75 - x_f} \right) = \frac{1}{1.4} \ln \left[\frac{0.4 x_f}{0.6(1-x_f)} \right] + \ln \left[\frac{0.4}{1-x_f} \right] \rightarrow e^{\ln \left(\frac{0.15}{0.75-x_f} \right)} = e^{\ln \left(\frac{0.4 x_f}{0.6(1-x_f)} \right)^{1/1.4} \cdot \frac{0.4}{1-x_f}}$$

Input values for x_f that converges to equal values

$$x_f = 0.5$$

$$\frac{0.15}{0.75-x_f} = \frac{0.4 x_f}{0.6(1-x_f)} \left(\frac{0.4}{1-x_f} \right)^{1/1.4}$$

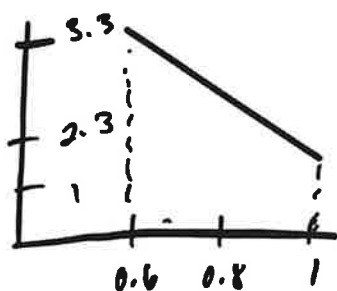
$$W_f = F \cdot \frac{x_{D,avg} - x_F}{x_{D,avg} - x_f} = 2 \cdot \frac{0.75 - 0.6}{0.75 - 0.5} = 1.2 \text{ kmol}$$

D11. A constant-mole batch distillation is used to change from a pure n-butanol solvent to a solvent that is 60.0 mol% n-butanol and 40.0 mol% water. If the initial charge is $W = 2.0$ kmol, how much water, S , must be added to achieve the desired solvent concentration? Do the integration with Simpson's rule, dividing the entire integration from 0.6 to 1.0 first into one step and then into two steps. Compare the answers.

$$x_F = 0.6 \quad \frac{1.0 + 0.6}{2} = 0.8$$

$$x_f = 1.0$$

$$W = 2.0 \text{ kmol}$$



x_D	x_W	y	$\frac{1}{y}$
1	0	1	1
0.8	0.2	0.435	2.3
0.6	0.4	0.3	3.33

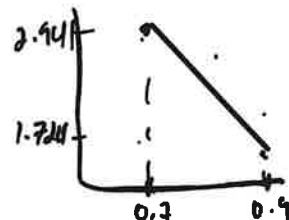
$$\text{Area} = \frac{1 - 0.6}{6} [3.33 + 4(2.3) + 1] = 0.902$$

$$\frac{S}{W} = 0.902$$

$$W = 2.0$$

$$S = 0.902(2.0) = 1.804 \text{ kmol}$$

x_D	x_W	y	$\frac{1}{y}$
0.9	0.1	0.58	1.724
0.7	0.3	0.34	2.941



$$Area = \frac{1-0.8}{6} \{ 2.3 + 4(1.724) + 1 \}$$

$$Area = \frac{0.8-0.6}{6} \{ 3.33 + 4(2.94) + 2.3 \} = 0.34$$

$$total\ Area = 0.92$$

$$\frac{S}{W} = 0.92$$

$$S = 2 \cdot 0.92 = 1.84 \text{ Knol}$$

operated with a reflux ratio that is fixed constant. We desire a still pot composition of 0.10 mol% benzene, and distillate should be 40% benzene. Pressure is 1 atm and (AM) is valid.

a. When the external reflux ratio L/D must be used?

b. What final external reflux ratio must be used?

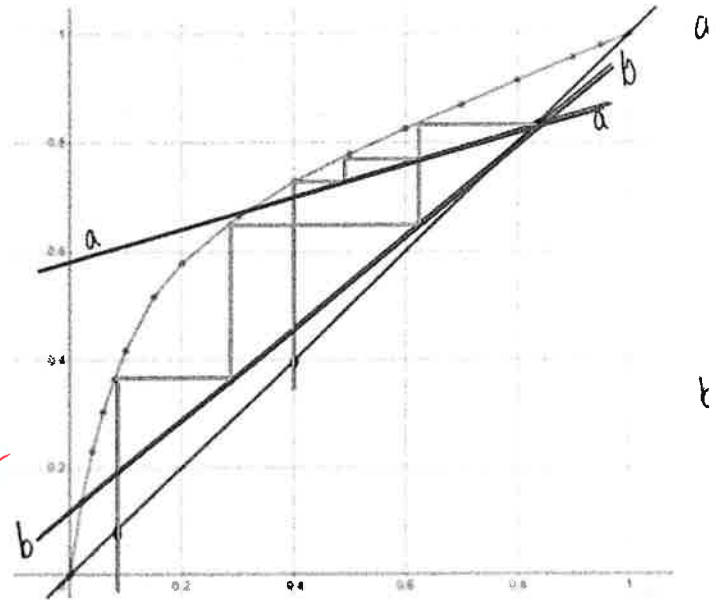
c. How much distillate product is to be drawn, and what is the value of W_f ?

$F = 10 \text{ kmol}$
 $X_F = 0.10$
 $X_D = 0.84$
 $X_f = 0.08$
 a) $\frac{L}{D} = ?$

$$\frac{L}{V} = \frac{0.84 - 0.56}{0.84 - 0} = 0.33$$

$$\frac{L}{D} = \frac{\frac{L}{V}}{1 - \frac{L}{V}} = \frac{0.33}{1 - 0.33}$$

$$\frac{L}{D} = 0.49 \quad 0.47$$



b) $X_f = 0.08$ $X_D = 0.84$

$$\frac{L}{V} = \frac{0.84 - 0.13}{0.84} = 0.845$$

$$\frac{L}{D} = \frac{0.845}{1 - 0.845} = 5.45$$

c) $F = D_{tot} + W_f$

$$10 = D_{tot} + W_f$$

$$FX_F = D_{tot} \cdot X_D + W_f \cdot X_f$$

$$(10)(0.1) = 0.84D + 0.08W_f$$

$$4 = 0.84D + 0.08W_f$$

$$4 = 0.84D + 0.08(10 - D)$$

$$4 = 0.84D + 0.8 - 0.08D$$

$$W_f = 10 - D$$

$$W_f = 10 - 4.21$$

$$W_f = 5.79 \text{ kmol}$$

$$3.2 = 0.76D$$

$$D = 4.21 \text{ kmol}$$